

## Project Initialization and Planning Phase

|               |                                                |
|---------------|------------------------------------------------|
| Date          | 2 July 2026                                    |
| Team ID       |                                                |
| Project Title | <b>Human Development Index (HDI) Predictor</b> |
| Maximum Marks | 5 Marks                                        |

### Project Proposal (Proposed Solution) report:-

The proposed solution aims to predict a country's **Human Development Index (HDI) category** using **Machine Learning**. The system analyzes key development indicators such as **Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income (GNI) per Capita** to classify countries into **Very High, High, Medium, or Low human development categories**. This solution enables researchers, policymakers, and students to perform fast and reliable HDI assessments through an interactive Streamlit web application.

| Project Overview  |                                                                                                                                                                                                                                        |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Objective         | The primary objective is to develop a Machine Learning-based HDI Prediction System that accurately predicts the Human Development Index category using country development indicators, enabling quick and data-driven decision-making. |
| Scope             | The project focuses on collecting and processing HDI-related data, training a Machine Learning model, and deploying an interactive Streamlit application that predicts the HDI category for user-provided country indicators.          |
| Problem Statement |                                                                                                                                                                                                                                        |
| Description       | Traditional HDI analysis requires manual evaluation of multiple indicators and interpretation of statistical reports. This process is time-consuming and does not provide instant predictions for researchers and policymakers.        |

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| Impact                   | The proposed system simplifies HDI prediction by providing fast, accurate, and automated classification, supporting policy planning, educational research, and development analysis.                                                                                                                                                                                                                                    |
| <b>Proposed Solution</b> |                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Approach                 | Develop a Machine Learning model using historical HDI data to analyze development indicators and predict the corresponding HDI category. The trained model is integrated into a Streamlit web application for real-time predictions.                                                                                                                                                                                    |
| Key Features             | <ul style="list-style-type: none"> <li>• Machine Learning-based HDI category prediction.</li> <li>• Interactive Streamlit web application.</li> <li>• Real-time prediction using Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per Capita.</li> <li>• User-friendly interface for researchers, policymakers, and students.</li> <li>• Fast and accurate prediction results.</li> </ul> |

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|  | <ul style="list-style-type: none"> <li>- Real-time decision-making for quicker loan approvals.</li> <li>- Continuous learning to adapt to evolving financial landscapes.</li> </ul> |
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## Resource Requirements

| Resource Type           | Description                             | Specification/Allocation                                                                                                                                                     |
|-------------------------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hardware</b>         |                                         |                                                                                                                                                                              |
| Computing Resources     | CPU/GPU specifications, number of cores | Apple M4 Processor (8-Core CPU), GPU not required                                                                                                                            |
| Memory                  | RAM specifications                      | 16 GB Unified Memory                                                                                                                                                         |
| Storage                 | Disk space for data, models, and logs   | 256 GB SSD                                                                                                                                                                   |
| <b>Software</b>         |                                         |                                                                                                                                                                              |
| Frameworks              | Python frameworks                       | Streamlit                                                                                                                                                                    |
| Libraries               | Additional libraries                    | scikit-learn, pandas, numpy, joblib, streamlit                                                                                                                               |
| Development Environment | IDE                                     | Visual Studio Code (VS Code), Python 3.13                                                                                                                                    |
| <b>Data</b>             |                                         |                                                                                                                                                                              |
| Data                    | Source, size, format                    | HDI.csv dataset, CSV format, approximately 190+ country records with Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, GNI per Capita, and HDI Category |